Lab 1 Submission

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Problem 1 Answer

Output:

```
det    Determinant.
    det(X) is the determinant of the square matrix X.

    Use COND instead of det to test for matrix singularity.

    See also cond.

    Documentation for det
```

Problem 2 Answer

```
C = [1:20]
C2 = C .^ 2
```

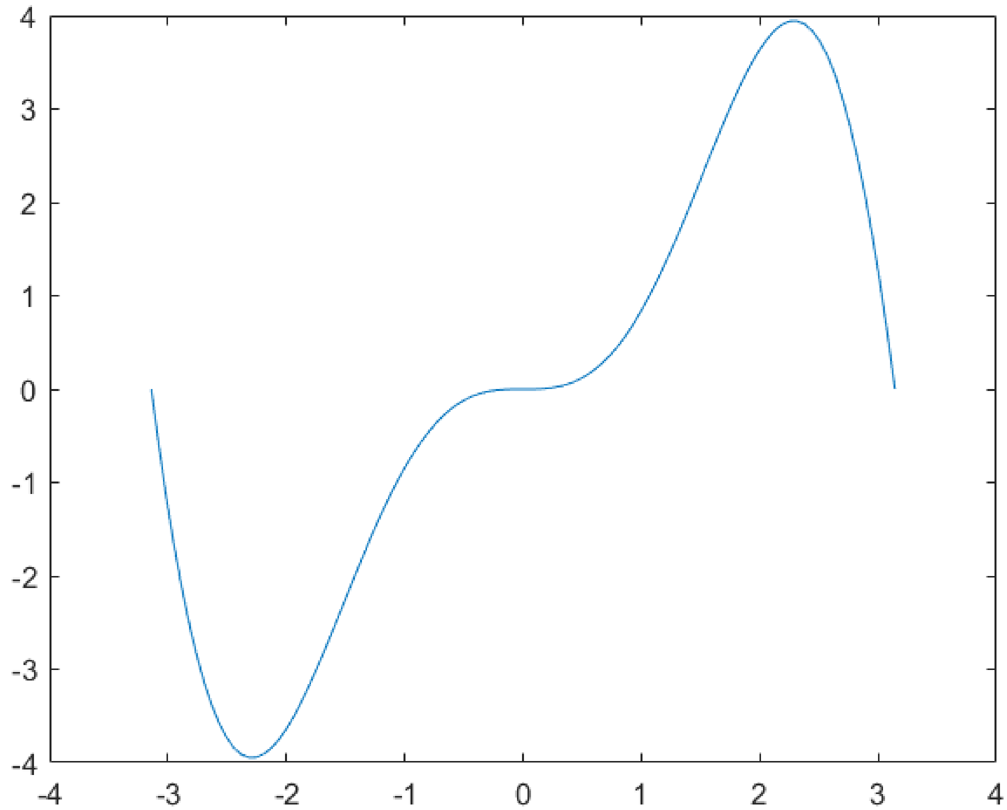
Output:

```

  Columns 1 through 14

     1     4     9    16    25    36    49    64    81   100   121   144   169   196

  Columns 15 through 20

   225   256   289   324   361   400
```

Problem 3 Answer

```
x = linspace(-pi,pi);
y1 = (x.*x).*sin(x);

figure
plot(x,y1)
```

Output:



Problem 4 Answer

```
function [result] = f(x)
    cumulative = 1
    for i = [1:x]
        cumulative = cumulative * i
    end
    result = cumulative;
end
```

Inputting x = 10:

```
ans =

     3628800
```

Which is the same as using the built-in factorial function

Problem 5 Answer

```
MyBisectionMethod(-1,3,0.00001,@(x) (x^2)-7)
function [result] = MyBisectionMethod(a,b,tol,f)
arguments
    a double
```

```
    b double
    tol double
    f function_handle
end
    cs = [];
    accuracy = 999;
    while accuracy > tol
        midpoint = (b+a)/2;
        cs = [cs, midpoint];
        if f(a)*f(midpoint) < 0
            b = (a+b)/2;
        elseif f(a)*f(midpoint) > 0
            a = (a+b)/2;
        else
            break
        end
        accuracy = (b-a)/2;
    end
    result = cs;
end
```

When run for the specified range, function, and tolerence:

```
ans =

  Columns 1 through 10

    1.0000    2.0000    2.5000    2.7500    2.6250    2.6875    2.6562    2.6406
2.6484    2.6445

  Columns 11 through 18

    2.6465    2.6455    2.6460    2.6458    2.6456    2.6457    2.6457    2.6457
```

Which means it took 18 steps to get to the specified tolerence

This is exactly as was predicted, since -1+log2((b-a)/tol) = ~17.6, which implies 18 subdivisions of the interval